

Virtual Private Network (VPN) Implementation

1. Introduction

The purpose of this lab is to investigate the mechanisms by which a Virtual Private Network (VPN) secures data and protects user privacy. As internet traffic traverses public infrastructure, it is susceptible to interception and tracking. This report documents the successful implementation of a VPN tunnel using ProtonVPN on a Windows environment to mitigate these risks.

2. Technical Theory & Key Concepts

2.1 The VPN Tunnel

A VPN functions by encapsulating standard data packets (like HTTP or DNS) inside an encrypted "outer" packet. This process is known as **Tunneling**. Even if a malicious actor intercepts the data on a public Wi-Fi network, they cannot view the contents of the inner packet.

2.2 Encryption: AES-256

ProtonVPN utilizes **AES-256 (Advanced Encryption Standard)**. This is a symmetric block cipher that uses a 256-bit key. It is mathematically impossible to crack with current computing

power, providing a secure barrier against Man-in-the-Middle (MITM) attacks.

2.3 Privacy: IP Masking

Every device on the internet has a unique IP address that reveals its location and ISP. By routing traffic through a VPN server, the destination website sees the **VPN Server's IP** instead of the user's real IP, effectively anonymizing the user.

3. Implementation Methodology (Steps)

Step 1: Installation The ProtonVPN Windows client was downloaded and installed. This software acts as the endpoint controller for the encrypted tunnel.

Step 2: Authentication The user authenticated via a secure account. This step ensures that a unique set of encryption keys is generated for the session.

Step 3: Protocol Selection For this lab, the **WireGuard** protocol was selected (or Smart Protocol) to ensure high-speed performance without sacrificing cryptographic strength.

Step 4: Connection Establishment The "Quick Connect" feature was activated. This initiated the TCP/UDP handshake with a remote server in [Mention the Country you connected to, e.g., Netherlands].

4. Verification & Results (Deliverables)

4.1 Evidence of Secure Connection

Upon connection, the ProtonVPN dashboard displayed a green "Connected" status, indicating that the encryption tunnel is active and all outgoing traffic is now protected.

[INSERT SCREENSHOT HERE: "Connection Status" - Show the ProtonVPN dashboard saying CONNECTED]

4.2 Comparative Analysis (IP Leak Test)

To verify the VPN was successfully masking the network

Metric	Before VPN	After VPN (Secure)
IP Address	[Your Real IP]	[The VPN IP from your Screenshot]
Location	[Your City]	[The VPN City, e.g., Amsterdam]
ISP	[Your Service Provider]	Proton AG / M247

identity, a check was performed using an external IP lookup tool.

5. Security Analysis & Observations

During the lab, the following security benefits were observed:

1. **Network Obfuscation:** The Internet Service Provider (ISP) can no longer track the specific websites visited. They only see encrypted traffic traveling to a single IP address (the VPN server).
2. **Public Wi-Fi Resilience:** In a real-world scenario, this setup prevents "Packet Sniffing" by attackers on the same local network.
3. **Kill Switch Utility:** (Optional Observation) If the VPN connection drops, the software's "Kill Switch" stops all internet traffic to prevent the real IP from accidentally leaking.

6. Conclusion

The objective of this project was successfully achieved. By implementing ProtonVPN on Windows, we established a secure communication channel that provides both **data integrity** (via encryption) and **identity protection** (via IP masking). This implementation is a critical first line of defense in modern network security.